

Consulting Proposal for Nicholas Vanderschantz

Kei Hea a Nic? Where is Nic?: Dynamic Digital Office Display System

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Table of Contents

Profile: Nicholas Vanderschantz and CMS 4
The Problem 7
The Solution..... 8
 Benefits..... 9
Project Schedule..... 12



Executive Summary

Academic staff within the School of Computing and Mathematical Sciences are highly collaborative and busy, and their availability changes frequently throughout the day. At present, staff communicate when they are free by displaying printed paper schedules outside their offices. While simple, these static schedules are difficult to keep accurate: as commitments shift, the printed information quickly becomes outdated, leaving students, colleagues and collaborators to rely on incorrect availability when trying to meet an academic and leaving staff with the ongoing effort of reprinting and replacing sheets.

This project proposes to replace that paper-based approach with a dynamic, easily updated digital display mounted outside the office. The proposed prototype consists of three integrated components:

- Diary Owner Web Application - a web-based interface that lets an academic manage and update the availability information shown outside their office.
- Diary Server - a backend service that securely transfers and manages availability data between the web application and the display device.
- Digital Display Interface - a Raspberry Pi-powered visualisation that presents up-to-date availability on a dedicated office screen.

Together, these components allow an academic to control what appears on their door in real time, so the displayed information stays current without manual reprinting. The expected outcomes are a reduction in the administrative effort of maintaining schedules, improved accuracy of the availability shown, and a clearer, more reliable picture for students and staff of when an academic can be reached.

The scope of the work spans early requirements analysis and definition through to paper, low-fidelity and high-fidelity prototyping, culminating in the three interfacing technologies described above. The school will supply the Raspberry Pi and screen, along with access to 3D printing for a suitable door-mounting solution; the team will design the webserver and networking components to operate within the provisions and constraints of the University of Waikato's IT environment.

By delivering this solution, the project aims to modernise a small but genuine everyday problem within SCMS turning a static paper sheet into a live, interactive display that improves communication and coordination across the school.

Title: Nicholas Vanderschantz

Academic staff in the School of Computing and Mathematical Sciences are highly collaborative and busy, with weekly diaries full of interactions with colleagues, collaborators, supervision of students and undergraduates. This makes finding a free moment to meet them genuinely difficult.

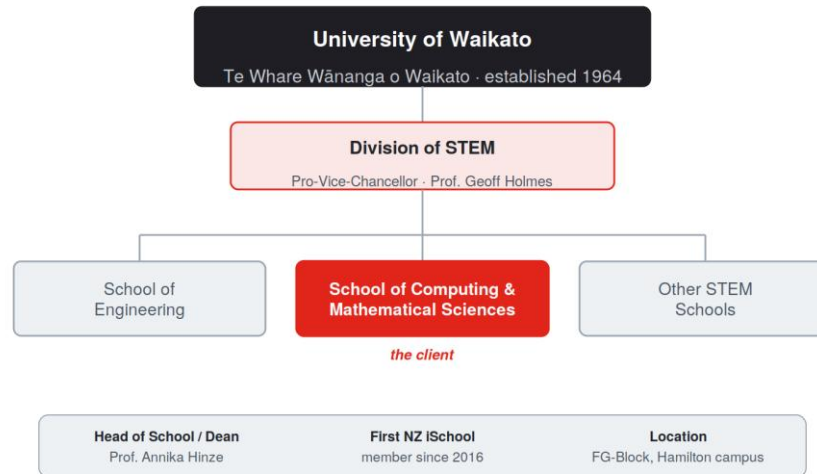
The school's current solution is for staff to advertise their basic availability on their office doors, typically as a printed paper sheet. This approach is static and quickly outdated: it cannot reflect same-day changes, offers no interactivity, and must be manually reprinted whenever a schedule shifts.

The problem this project addresses are therefore how to replace that printed sheet with a dynamic, interactive and easily accessible digital display of an academic's availability, driven by a screen and Raspberry Pi mounted at the office door and kept current by the staff member themselves.

Profile: Nicholas Vanderschantz and CMS

The client for this project is the School of Computing and Mathematical Sciences (SCMS), one of the teaching-and-research schools of the University of Waikato. The immediate stakeholders and end-users are the academic staff of the school, the professors, lecturers, teaching fellows and postgraduate researchers whose office doors currently carry the printed availability sheets this project sets out to replace. They are, in the words of the project brief, "collaborative, social, and busy" individuals, and it is their day-to-day scheduling problem that defines the need.

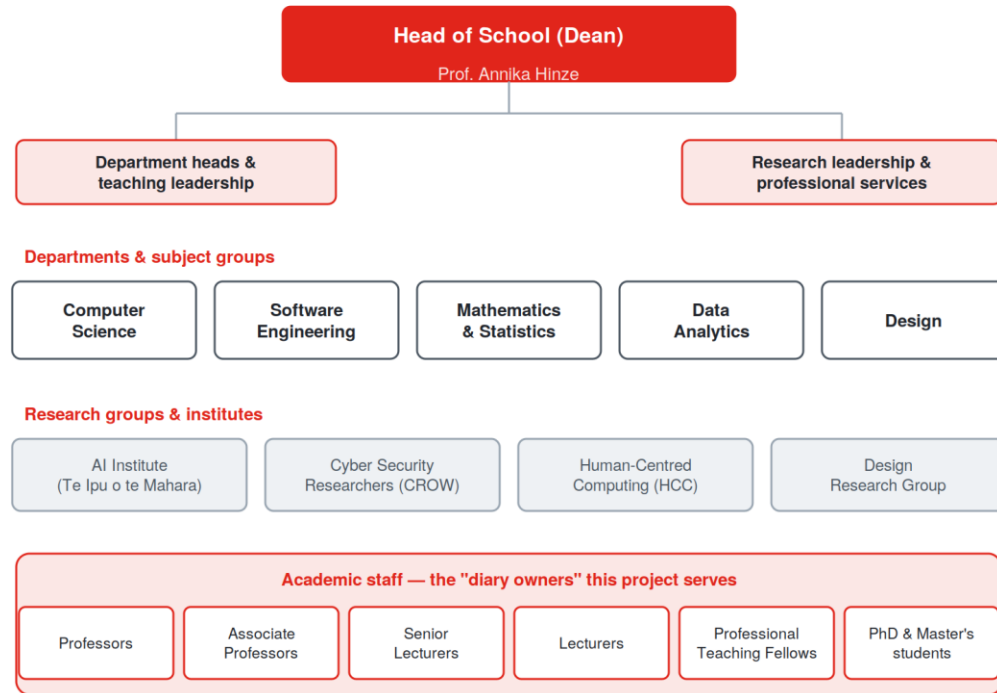
The School of Computing and Mathematical Sciences sits within the University's Division of STEM and within that division, SCMS is one of several schools, alongside the School of Engineering and other



The University of Waikato educates approximately 13,000 students and employs around 1,500 staff across its campuses. The School of Computing and Mathematical Sciences is a sizeable part of this. It delivers undergraduate and postgraduate programmes across five broad subject areas and hosts several research groups and institutes.

SCMS combines teaching, research, and industry engagement across computing, mathematics, and design. It offers three and four-year undergraduate degrees built on a common foundation in computing and mathematical sciences, together with postgraduate and doctoral study.

Internally, the School is led by the Head of School and organized into departments and subject groups, overlaid with research groups and supported by professional services staff.



A SCMS academic's working week is unusually heavy interaction. Alongside preparing and delivering lectures and labs for undergraduate cohorts, staff supervise PhD and master's students, collaborate on research with colleagues inside and outside the University, sit on committees and carry administrative duties, and engage with industry partners. Each of these threads generates meetings, consultations and drop-ins, and each pull on a different, shifting part of the diary.

Because so many people need time with each academic, and because that availability changes constantly, finding a moment to meet is genuinely difficult, which is precisely the problem this project addresses.

Within the School it is common for staff to advertise their basic availability publicly on their office doors, most often as a printed paper timetable pinned or taped to the door. This is simple and glanceable, but static: it dates quickly, cannot reflect same-day changes, and offers no interactivity. As computer scientists and designers, the client asked whether modern digital-display technology could do better.

This project responds to that question by proposing a dynamic, interactive, and easily accessible digital display. The solution is envisaged as three interfacing technologies: a Diary Owner App, a Diary Server, and a Diary Visualization Interface.

In short, the client is both the beneficiary of the system and the setting whose people, structure and working patterns shape its requirements.

The Problem

Academics within the School of Computing and Mathematical Sciences (SCMS) frequently interact with a wide range of people, including undergraduate students, postgraduate students, research collaborators, colleagues, and university staff. As a result, academics need an effective way to communicate their availability for meetings, discussions, supervision sessions, and other forms of interaction.

Currently, many academics communicate their availability by displaying a printed schedule outside their office door. These schedules typically show planned availability, teaching commitments, meetings, and periods where the academic is unavailable. However, printed schedules only represent a fixed snapshot of an academic's planned schedule and rely on manual updates when changes occur.

Academic schedules are often dynamic and can change due to meetings, teaching responsibilities, research activities, supervision commitments, administrative tasks, and unexpected events. Because printed schedules do not automatically reflect these changes, the displayed information can quickly become outdated. As a result, students and colleagues may make decisions based on availability information that is no longer accurate.

The current format also limits the amount of information that can be communicated. A printed schedule provides only a planned overview and cannot easily represent changes, additional context, or circumstances that occur after it has been displayed. Furthermore, because the schedule is located only outside an academic's office, individuals must physically visit the office to access the information and have no reliable way to determine whether the displayed schedule is still current before doing so.

This issue affects a broad range of SCMS stakeholders, including undergraduate students seeking assistance, postgraduate students meeting with supervisors, academic staff coordinating research or teaching activities, administrative staff, and external collaborators. Across multiple academic offices and frequent daily interactions, inaccurate availability information can lead to unnecessary office visits, delays in communication, and additional effort spent determining when academics are available.

Although the current approach is simple and familiar, its limitations create ongoing challenges in coordinating interactions between academics and the many individuals who rely on accurate availability information. The issue is not limited to individual instances of inconvenience; across SCMS, repeated reliance on outdated schedules can create inefficiencies in everyday academic communication and collaboration.

The Solution

The proposed solution is a prototype for a digital office availability system that replaces printed schedules with a dynamic, web-based solution that can display availability information in real time. Rather than relying on a printed timetable attached to office doors, the system will allow academics to easily update the information displayed through a simple web application. Any changes made by the academics will be automatically updated on a dedicated digital display, ensuring that visitors, colleagues and students will have access to the most up-to-date and correct information.

The prototype consists of 3 different components that communicate with each other.

Web Application – A secure web application that allows the academics to manage and view the information that will be displayed on the display. They will be able to update their current availability, office hours, meeting status, expected return time, and any custom messages that they wish to display.

Server - A server that will act as the bridge between the web application and the display. The server receives updates from the web application and stores the current availability and the timetable. And sends the data to the display component.

Digital Office Display Interface – A full-screen web application runs on the Raspberry Pi connected to a monitor that is outside the academic's office. The display presents the academic's current status in a clear, concise and easy-to-read format and updates automatically whenever changes are made through the web application. Visitors will be able to use the display system to see the academic's current availability, timetable and any messages.

The prototype will use modern web technologies to demonstrate how digital displays can provide a more flexible alternative to traditional paper schedules. A database will store availability information and configuration settings, as well as the calendar information used by the system, while the server will manage the communications between the components. The Raspberry Pi will continuously get new data and display the latest up-to-date information.

Due to the project being a prototype, it will focus on demonstrating the feasibility rather than providing a complete production-ready system. The prototype will allow academics to update their availability manually through the web application and have those changes near-instantly updated on the digital display. Additionally, features such as a standalone calendar, calendar event importing from existing calendars via .ics files, a QR code that allows visitors to access an appointment request form, and authentication will be explored as part of the prototype. The exact implementation of the appointment request form will be decided during the design and development stage of the project.

The main objective of the system is to provide academics with an easy way to manage their availability while providing visitors, colleagues or students with accurate and accessible information about when an academic is available.

Benefits

Real time accurate availability:

Paper schedules printed and put on doors are static. They are often out of date or unreliable as availability changes day to day in the fast-paced environment that the University of Waikato presents. The use of a digital system allows for more accurate availability in real time. Going digital removes the need for guesswork or assuming that the academics might be available at times when they might be busy. This would reduce the number of wasted trips students and colleagues make to classes and offices in search for the academics when they are somewhere else. Accuracy and adaptability are key strengths of transitioning to a digital system and are expected to provide meaningful time savings for both students and academics by reducing unnecessary trips, delays, and uncertainty when determining availability.

Reduction in administrative overhead

Moving to a digital solution also means less work for academics to keep up with displaying their information. Paper schedules mean constant reprinting and hanging up of their schedules, which is a small but repetitive process that adds up over a semester. Our proposed system simplifies the process of making changes, allowing academics to quickly and easily update their availability information directly through the web application. Updates are automatically sent from the web app to the screen, removing much of the manual work academics previously had to carry out. Based on the reduction in manual steps required, we estimate that updating availability information could be approximately 90-100% faster than the current process of manually printing and replacing schedules. The digital process is expected to be substantially faster than manually printing and replacing schedules each time an update is required.

Faster, lower-friction meeting scheduling

As the digital solution will allow for better accuracy and always reflect current availability it will make it easier for colleagues and students to find a suitable meeting time. It will remove uncertainty around scheduling, saving academics from sending emails back and forth, making the whole process more efficient. The inclusion of a digital interface for students and academics to book a timeslot on the screen would further reduce the friction. Users can scan a QR code to get the academic's contact details, making the process more efficient for all users with everything displayed in one place.

Consistency and professional presentation

As mentioned by our stakeholder academics vary in the way they like to display their schedule and timetable. This can make it confusing for students and colleagues who are used to other layouts and schedules shown on other doors. For example, some people prefer to have their schedule show date horizontally and time vertically, whilst some have it vice versa. This can make it difficult for students and colleagues to quickly read and understand the schedule. A universal system allows for a more consistent and professional presentation that saves time for students and academics to catch information at a glance. Currently academics use handwritten notes, printed timetables and sticky notes with crossing out and different coloured pens to show changes making it hard to keep up with changes whilst staying presentable and professional.

Improved accessibility

The proposed digital system will also support larger, adjustable text, with higher contrast colour schemes and colour-blind settings. This would allow for easier and more inclusive reading where users won't have to squint to read the text on a tiny A4 schedule taped to the door. There will also be further accessibility features such as text-to-speech capabilities or braille support for visually impaired users. This would ensure that all students would be able to access and understand the schedule. In addition, a digital bright and colourful solution is more visually appealing and engaging for users compared to a simple black and white printed schedule which will help to attract attention to key information saving time for students scouring the schedule for information.

Environmental benefits

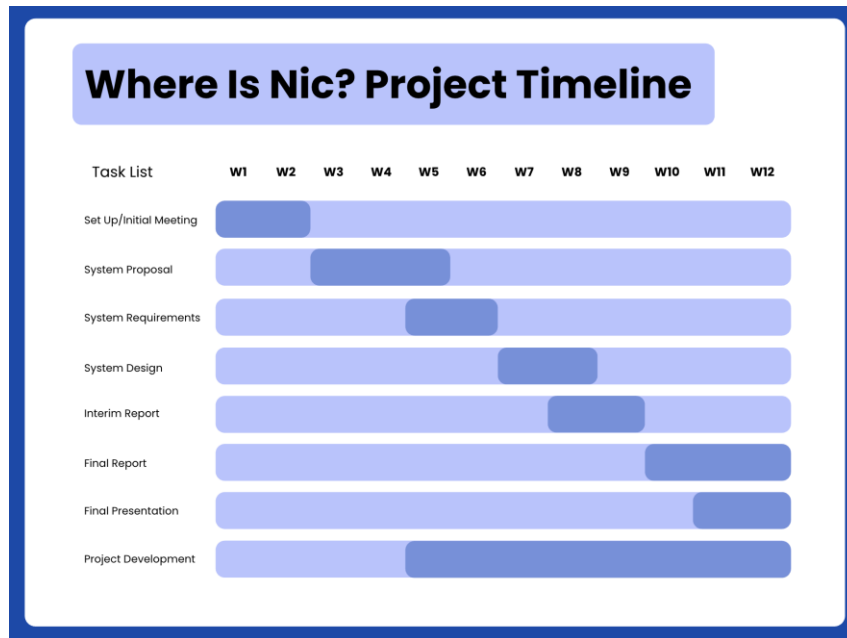
The proposed digital solution also offers meaningful environmental benefits by reducing the school's reliance on paper and ink. Under the current system, printed schedules must be updated and reprinted by academic staff whenever their availability changes, alongside additional handwritten notes and sticky notes used to convey short-term amendments. While the resource cost of a single printed page may appear negligible in isolation, this process is repeated by academic staff across the school on an ongoing basis throughout each semester, resulting in a cumulative volume of paper and ink consumption that is significant when considered at the scale of the whole School over an academic year. By displaying availability information digitally, the proposed system removes this recurring need for printing entirely, reducing both paper waste and the associated material costs, while supporting the school's broader efforts toward more sustainable operating practices.

Foundation for improvement

The proposed digital system is built on modern web technologies it can be further developed and improved upon in the future. This means later down the line the system could include features like integration with the actual school timetables on MyWaikato or external calendar platforms like Google Calendar etc. This allows for the school to be more flexible and adaptable to the needs to the students and academics-a-like versus a one-off fix that the paper system provides.



Project Schedule



System Proposal: 27/7 - 15/8

During the beginning phase of the project, we will focus on understanding the current state of the office availability system in the school and identify how a digital system could improve upon it. We will conduct initial interviews with Nic, our client, to gather base layers of information that we can then use to inform the design of the system. We will then use this information to enable us to understand the full scope of the project, so we can work out what we need to do to meet the client's needs. We will also begin to work as a team for the first time, developing our communication through services like Discord and group meetings. We will mainly use this time to get to know each other and the project so we can build strong foundation for the rest of the project. By the end of this period, we will have a finalised our proposal for the project and a good comprehension of the client's needs.

System Requirements: 10/8 - 21/8

The next phase of the project will be shifting our understanding of the project into defining the functional requirements of the system we must meet to fulfill the client's criteria. We will continue to conduct our own group meetings and meetings with Nic to gain a more in-depth understanding of the project and the problem the client is trying to solve. We will make decisions on key features and functions of the system during these meetings with Nic to set us on the right path for the scope of the project. By the end of this phase, we will have a finalised set of functional requirements for the system that our project must meet to be a practical and complete solution for the client.

System Design: 7/9 - 18/9

Utilising our system requirements document we will be able to develop various prototypes and wireframes for our system. We will need to create and consider multiple different design styles and layouts for our system which we will consult with Nic. This will require frequent meetings with Nic to ensure we are meeting his needs and requirements for functionality and design. We will also be able to utilise Nic's expertise in the Design field to ensure our system will be well developed and thought out. Upon completion of the System Design phase, we will have fully explored the use of interface and database design for our system. We will have assessed the feasibility of our system and ensured that we understand how clients and users will interact and use our system.

Project Development: 10/8 - 16/10

Once we have completed all our deliverables, we will then be able to move on to developing our prototype. We will develop a full complete prototype of our system that will aim to meet all the functional requirements we set out in our second deliverable. We will be working collaboratively as a group to develop our prototype together using a shared GitHub repository. We will need to create three integrated components to develop our solution: A diary owner web application, a diary server, and a digital display interface. We will continuously update and assess our prototype as we develop it to ensure it is meeting both the functional and non-functional requirements ready to be handed over to the client upon completion.

Final Presentations and Reports: 14/9 - 16/10

While we are developing our final system we will also need to be considering how we are going to present it and document our work. Alongside our final prototype we will present a thorough report of our work and development expressing challenges we faced along the way. We would have completed most of the development for the prototype at this stage and have enough to get a picture of the outcome to begin to form a complete presentation and report. The report and presentation will enable the client to have a better understanding behind our thought process for every decision and development process. The documentation will act as invaluable resource for the client to then further develop and enhance the prototype in the future, providing key detail and technical explanation.